

Problem Set 7

POLS W4732

Fall 2026

Due December 4 before the TA section

Instructions. For each formal claim, include two or three algebraic inequalities or best-response comparisons that establish it. Unless directly following notation from slides or the text of the problem, define your notation.

Artificial-intelligence policy. Students may not use generative-AI systems (e.g., ChatGPT, Claude, Gemini, Copilot, etc.) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, and collaboration within the registered problem-set group are permitted. **Every group member is responsible for understanding and being able to explain the entire submission.**

Exercise 1. Consider a society composed of three groups: R , M , and P . In all groups, there is a unit 1 of voters (they are equally sized). The utility of each group depends on public spending on two policies: p and s (e.g., police and social security). Each group differs on how much they value policy s . The utility of group $J \in \{R, M, P\}$ is:

$$U_J(p, s) = p(\alpha - p) + \beta_J s \tag{1}$$

Assume that $\beta_R = 0$, $\beta_P = 1$ and $\beta_M \in (0, 1)$, and $\alpha \in (1, 2)$.

Spending on both policies cannot go over the (exogenous) budget 1: $p + s \leq 1$. Two candidates A and B compete to get elected. They commit to a platform (p_i, s_i) , $i \in \{A, B\}$ before the election. After observing candidates' platforms, voters vote for one of the two candidates.

1. Find the ideal spending mix on p and s for each group, denoted by $p^*(\beta)$.
2. Show that under perfect information and deterministic voting (each voter votes for the candidate whose platform yields the highest payoff), both candidates propose group M 's preferred spending mix.
3. Substitute the budget constraint into each voter's utility and write down $V_J(p)$, the indirect utility of a voter from group j of a platform p .

4. In what follows, assume that candidate A and B are both policy-motivated. Candidate A has the same preference as voters' from group R ; candidate B has the same preference as voters' from group P . Furthermore, assume that total turnout in groups M and P is distributed uniformly on $[0, 2]$. That is, there is a probability $\frac{x}{2}$ that $x/2$ voters from group M and P (so x voters total) show up to vote. All voters from group R always vote.

Show that convergence is not an equilibrium outcome.

5. Suppose B proposes $p_B = p^*(\beta^P)$. Find the set of platforms which guarantee victory for A (assume that when indifferent, voters from group M vote for A so you can define a closed interval).

6. Now suppose that all voters turn out, but that voters in each group vote for candidate B if and only if

$$V_J(p^A) \leq V_J(p^B) + \xi$$

where ξ is distributed uniformly in $\left[-\frac{1}{2\psi}, \frac{1}{2\psi}\right]$.

Write down candidate A 's winning probability as a function of the platforms: $\pi(p^A, p^B)$.

7. Write down the objective function of each party, and the associated first-order conditions.
8. Using the first order conditions, argue that $P_A(p_B)$, A 's best reply to p_B is increasing and strictly larger than p_M^* , and that $P_B(p_A)$, B 's best reply to p_A is increasing and strictly smaller than p_M^* .
9. Suppose that $\alpha = 2$ and $\beta_M = 1/2$. Show that the distance between the equilibrium platforms $p_A - p_B$ decreases with ψ . [Hint: guess that $p_A + p_B = 3/2 = 2p_M^*$, and use the FONC to obtain a simple expression characterizing $\Delta = p_A - p_B$.]

Exercise 2. Consider the baseline version of the cheap talk model. Prove that the Sender is worse off in a babbling equilibrium than in a partially revealing equilibrium with two possible reports.